BC3415 · GROUP [number]
PROPOSAL TO CAROUSELL'S CTO AND CFO

1 **Business case**
Part 1 of 6

Carousell Dispute Agent

AI that helps support staff settle "not as described" disputes, with policy citations. A person approves every decision.

OUR RECOMMENDATION

APPROVE A STAFF-ASSIST PILOT

Annual value, base case	**S$400,000**
Value range	**S$71k – S$2.0M**
Payback on build (estimate)	**≈ 5 months**
Decision accuracy (50 test cases)	**88%**
Eligibility calls right	**100%**

Feasible today, at almost no running cost

A working prototype on a standard stack; production needs integration, not research

Architecture

Staff app
Streamlit: review, approve or override, what-if, policy library

Agent workflow
LangGraph, pauses for a person (interrupt)

AI calls
Google Gemini, structured JSON output, temperature 0

Policy search
ChromaDB embeddings + BM25 keywords, rank fusion

Unit economics and readiness

8.9 s
median time to a cited recommendation

1.8
AI calls per case (ineligible cases need none)

< S$0.01
AI cost per case; about S$50 a year in the base case

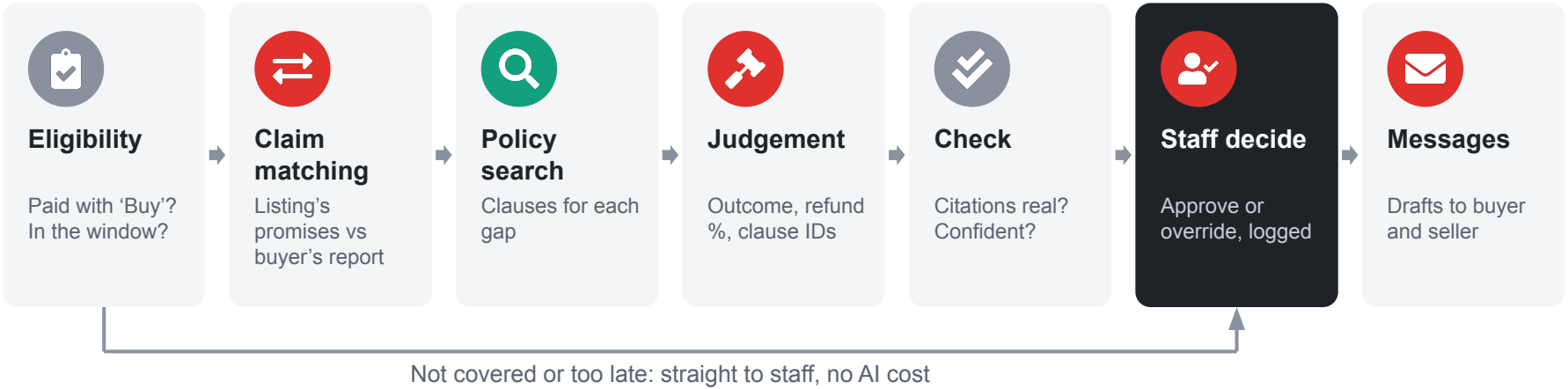
32
automated tests that run offline without an API key

Path to production: connect to Carousell's case system, remove personal details before any AI call, and move to a paid API tier so prompts aren't used for training.

Sources: reports/evaluation_gemini_test_v2.md (time and calls per case); AI cost per case from third-party-listed Gemini Flash-Lite prices (report 9.6, check Google's pricing page); 32 pytest tests in tests/.

Rules in code, judgement by AI, a person decides

Each step does what it does best, and staff approve or override every decision



Exact rules never miss
eligibility 100% vs 74% for a single AI prompt

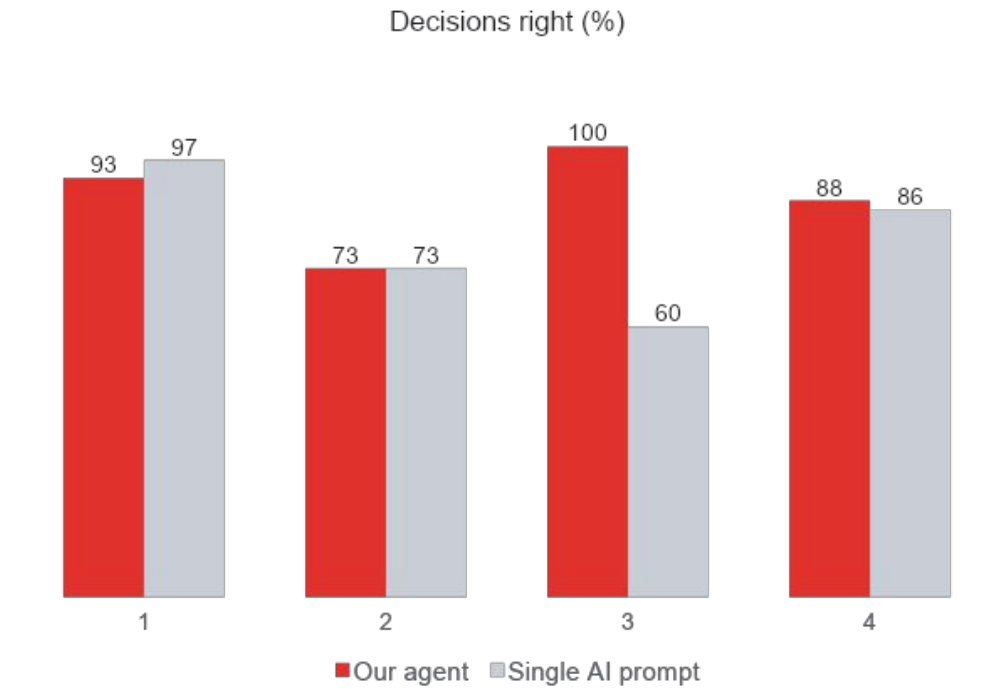
Unsure cases go to a person
6% escalated, including 2 of its 6 wrong answers

Labelled and checked
two of us hand-label the test set; an AI reviewer agreed on 47 of 50 (kappa 0.90)

Workflow: src/dispute_agent/agent/graph.py. Escalation if confidence < 0.7, no valid citation, more evidence needed, an unclear gap, or a self-contradicting outcome. Labels: report section 7.2.

Level on judgement, clearly ahead on the rules

Against a single AI prompt with the whole policy, it wins where errors are costly



100% vs 74%

eligibility right: never misses a bank transfer, a late dispute or a completed order

92% vs 88%

right on the 94% of cases it doesn't escalate: the ones staff can approve quickly

82% → 88%

our first run lost; we fixed a contradiction in our decision guide on tuning cases only, then re-ran once

Source: reports/evaluation_gemini_test_v2.md, 50 held-out cases, judge gemini-flash-lite-latest. Labels are provisional until hand labelling is finished. One case = 2 points.

Worth about S$400,000 a year in Singapore

Staff time covers part of the cost; a small trust effect on ‘Buy’ covers the rest

Annual value (S$), by scenario

	Low	Base	High
Disputes reaching staff a year	1,440	**6,000**	28,000
Staff time saved	S$8k	**S$80k**	S$756k
More fees on ‘Buy’ (trust)	S$63k	**S$320k**	S$1.2M
Annual value	**S$71k**	**S$400k**	**S$2.0M**

Key drivers: each 1 point of deals moving onto ‘Buy’ ≈ S$320k in fees; one escalated dispute ≈ S$27 of staff time (45 min at S$35/h).

Cost and payback (estimate)

One-off build and integration	**S$150k**
Upkeep a year	**S$50k**
AI usage a year	**≈ S$50**
Net value a year (base)	**≈ S$300k**
Payback on the build	**≈ 5 months**

Break-even: +0.06 pt

About 6 more ‘Buy’ deals in every 10,000 cover the yearly cost. That’s one-sixteenth of our base assumption.

Source: project Business Case (31 Oct 2026): inputs sourced or labelled as assumptions, each with low/base/high values. Costs, net value and payback: suggested estimate (report 9.6), not yet agreed.

Executive summary

A Carousell-specific agent: faster, fairer disputes that pay for themselves through trust

The problem	‘Buy’ only protects buyers who trust how disputes end; scam victims paid outside it. Disputes about used goods are judgement calls.
Why it’s unique	Built for used goods between individuals and Carousell’s own policy. Shopee’s engine and generic refund tools are built for new goods from businesses.
The solution	Rules in code, AI for judgement, Carousell’s policy clauses cited for every decision, and a person approving each one.
Feasibility	Working prototype: about 9 seconds and under S$0.01 of AI cost per case, 32 automated tests, standard Python stack.
The proof	On 50 held-out cases: 88% decisions right (single prompt 86%), 100% eligibility (74%), 0 invented clauses.
Business case	S$400k a year base case (S$71k–S$2.0M). Breaks even if 6 in 10,000 more deals use ‘Buy’ (estimate).
Our ask	Approve a staff-assist pilot that measures time to resolve, decisions overturned and ‘Buy’ usage.

Trust in ‘Buy’ is the revenue lever

Scam victims paid outside the protected ‘Buy’ button; fair, fast disputes keep buyers inside it

Scams happen outside Buyer Protection

29% of Singapore’s 6,703 e-commerce scam cases in 2025 were on Carousell; victims paid by Pay Now or bank transfer

1,555 Carousell scam cases in H1 2026, up 17.1%: the only designated online service to rise

78,000+ user reports Carousell’s trust team reviewed in Singapore in 2025

Why disputes are hard to settle

COVERED

- Wrong model or colour
- Missing promised parts
- Undisclosed damage

NOT COVERED

- “Light pink” vs “rose pink”
- Disclosed wear
- Change of mind

Both kinds of error cost Carousell

Too pro-buyer: honest sellers lose money and leave. Too pro-seller: buyers stop using ‘Buy’ and pay by transfer, where scams happen, and Carousell loses the fee.

Sources: SPF Annual Scams and Cybercrime Brief 2025; SPF Mid-Year Brief 2026; Carousell Regional Scam Trends Report 2025; Carousell SG help centre (copied 28 Oct 2026).

No existing option fits Carousell's disputes

Other tools serve new goods sold by businesses; Carousell's disputes are over used goods

Option	Built for used goods between individuals	Available to Carousell	Fast decisions	Cites Carousell's own policy	Scales without more staff
Today's manual review	✓	✓	✗	–	✗
Shopee's AI return engine	✗	✗	✓	✗	✓
Generic AI refund tools	✗	✓	✓	✗	✓
Taobao public jury	✓	✗	–	✗	–
Our Dispute Agent	✓	✓	✓	✓	✓

Shopee: built for new goods from businesses, and not for sale

Generic tools: apply a shop's own return policy to its own stock

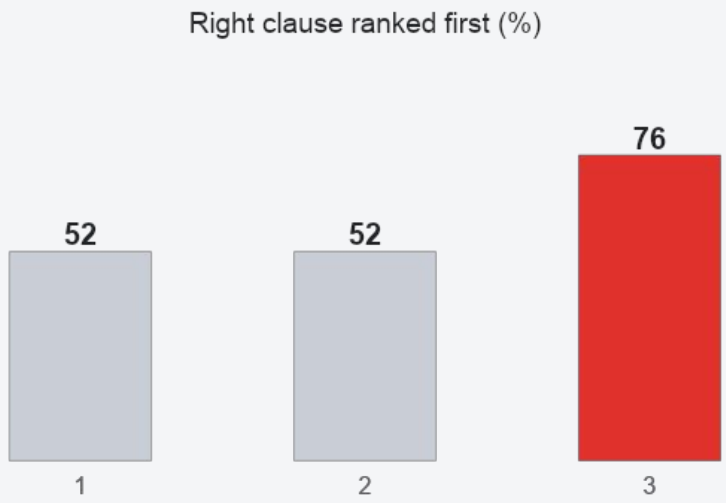
Taobao jury: needs a huge user base; seller jurors favour sellers

Sources: Sea Limited Q4 2024 earnings call (Shopee); vendor sites (generic AI refund tools, e.g. Fini); report section 4.6. "–" means no public information.

Grounded in Carousell's own policy

Every decision cites Carousell's own clauses, and code rejects anything not retrieved

Hybrid search finds the right clause more often



96% of the time the right clause is in the top five

Citations you can audit

0% of cases with an invented clause ID (single prompt: 4%)

91% of citations (64 of 70) support the decision they back up

37 clauses from 4 Carousell help-centre pages, each with a stable ID like BPR-03

Sources: reports/retrieval.md (25 test questions, 37 policy clauses from 4 Carousell SG help pages); reports/evaluation_gemini_test_v2.md; citation check by an AI reviewer, being confirmed by hand.

Test a policy change before it goes live

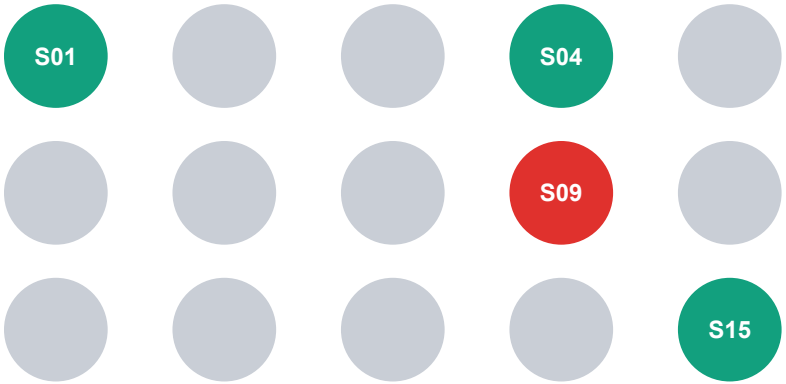
Re-judge past disputes under a proposed rule to see its impact before rollout

PROPOSED RULE

“If a promised accessory is missing, the buyer gets a full refund with return instead of a partial refund.”

- Re-judges past disputes with the new rule added as a clause
- Cheap: past results are cached, so each rule costs one AI call per case
- A person reviews every flip before the rule goes live

4 of 15 past decisions would change



- Intended: a promised accessory was missing
- Wrong flip: the listing never promised it
- No change

Source: what-if test on 15 seed cases (report section 8.4; src/dispute_agent/evaluation/whatif.py). AI answers can vary between runs.

Live demo: three real-looking disputes

Watch the agent apply the rules, cite the policy, and hand unsure cases to a person

1

S01 · PS Vita Slim

A promised part is missing

WATCH FOR

Partial refund 15%, citing the clauses on incomplete items; staff approve; messages drafted

2

S16 · Anime DVD

Paid by bank transfer

WATCH FOR

Rules say not covered straight away, with no AI cost

3

G155 · Nintendo games

The evidence conflicts

WATCH FOR

Confidence 0.5, so it goes to a person with a question for the buyer

Switching to the live app. Backups if an answer differs: S02 (iPhone 64GB received as 16GB, full refund with return), S10 (disclosed pilling, decline).

Tested on 200 realistic disputes

No public dispute data exists, so we built it on real listings and kept 50 cases aside

300

real listings

second-hand items from Mercari

200

disputes on top

20 by hand, 180 by a generator model

50

held-out test cases

never used to tune prompts

- **Realistic mix:** 17 scenarios in the planned 60/30/10 split of clear, borderline and ineligible cases, e.g. missing accessory, shade difference, paid outside ‘Buy’
- **Exact eligibility labels:** code sets the payment method and dates, so covered / not covered is certain
- **No model grades its own work:** the generator (Gemma 4, Gemini 3 Flash) is never the judge (Gemini Flash-Lite)
- **Labels checked:** provisional labels now; two of us label the test set by hand; AI reviewer agreed on 47 of 50

Listings: Mercari Price Suggestion dataset (Kaggle; US second-hand items, prices converted to S$). Disputes: data/cases/. Report sections 7.1–7.2.

Investment risks and mitigations

The main risks are known, measured on our test set, and handled by design

Risk	Why it matters	Mitigation and evidence
1 **Invented or wrong policy**	A decision cites a rule that doesn't exist and someone loses unfairly	Must cite retrieved clause IDs; code drops anything else and escalates. 0% invented (single prompt 4%)
2 **Bias between buyer and seller**	The AI favours whoever writes more persuasively	Accuracy tracked separately: 92% of seller wins vs 83% of buyer wins. Leans to declining; monitored in the pilot
3 **Staff rubber-stamping**	Staff approve without reading	Confidence shown, unsure cases escalated (6%), every override logged
4 **Privacy and data**	Real disputes contain names, addresses and chats	Prototype uses made-up data only; production removes personal details and uses a paid AI tier

Source: reports/evaluation_gemini_test_v2.md; report section 10. Buyer-win cases are those labelled as a refund; seller-win cases, a decline.

Our ask: approve a staff-assist pilot

A staged pilot proves the value with Carousell's own data before any wider rollout

1. Validate	2. Shadow mode	3. Staff-assist	4. Scale
Hand-label the test set; time staff with and without the agent	Agent recommends silently beside staff on real cases, personal details removed	Staff see recommendations and approve or override every case	Other Buyer Protection markets: HK, MY, TW, VN
2 weeks	**4 weeks**	**8 weeks**	**After review**

What the pilot measures

1. Time to resolve a dispute
2. Decisions overturned on appeal
3. Share of orders paid with 'Buy'

DECISION REQUESTED

Approve the pilot

Estimated S$150k build and S$50k a year to run, against S$400k a year of value in the base case.

Pilot stages and timings are our proposal. KPIs from report section 10; other markets from report section 9.3. Costs: estimate (report 9.6).

CAROUSELL DISPUTE AGENT

Thank you. Questions?

Code and reports:
github.com/J0knee10/carousell-dispute-agent-bc3415

1. Business case
2. Workflow and labelling
3. RAG and policy
4. App and deployment
5. Data and evaluation
6. Risks and recommendation